# Material Classes

14 May 2025 11:33

## Metals and Alloys

**Characterised by:**

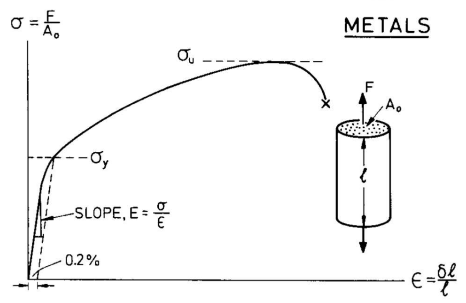
→ Relatively high moduli

→ Ductile, so can be formed

→ Even high strength alloys, with ductility as low as 2% yields enough to ensure failure is a tough ductile type

→ Partly because of ductility, prone to fatigue

→ Some have poor corrosion resistance




**Strengthened by:**

→ Alloying

→ Deformation

→ Heat treatment

## Ceramics & Glasses

**Characteristics:**

→ Typically high moduli

→ Brittle (No ductility)

→ Strength in tension = Brittle fracture strength

→ Strength in compression = Brittle crushing strength (~10x larger)

→ Low tolerance for stress concentrations (holes or cracks) or high contact stresses (at clamping points)

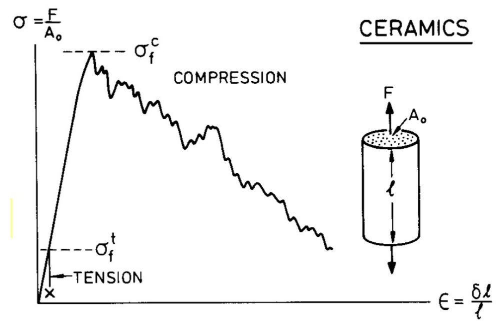
→ Scatter in strength depends upon volume of material under load

→ Harder to design with than metals

→ Abrasion resistant (Bearings and cutting tools)

→ High temperature capability

→ Corrosion resistant




## Composites

→ Combination of 2 materials to exploit most attractive properties

→ Typically polymer matrix (epoxy or polyester), usually reinforced by fibres such as glass, carbon fibre or Kevlar.

**Characteristics:**

- Light
- Stiff
- Strong
- Tough
- Polymer composites cannot be used above 250°C
- Expensive
- Difficult to form and join
- Used if added performance justifies added cost

## Polymers

**Characteristics**

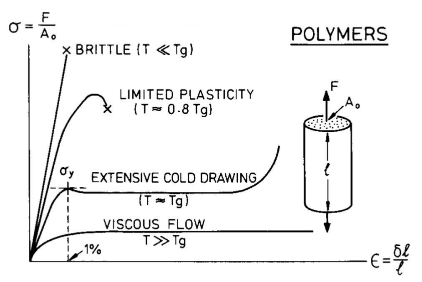
→ Low moduli (~1000x less than metals)

→ ~5x lighter than metals.

→ Some nearly as strong as metals

→ Large elastic deflections

→ Creep easily, even at room temp, so over time may reach permanent set if under load.




→ Properties depend on temperature

→ NONE have useful strength above 200°C

→ Easy to shape / form

→ Large elastic deflections allow for snap fixtures
   - Fast & Cheap assembly

→ No finishing operations needed if mould is accurate

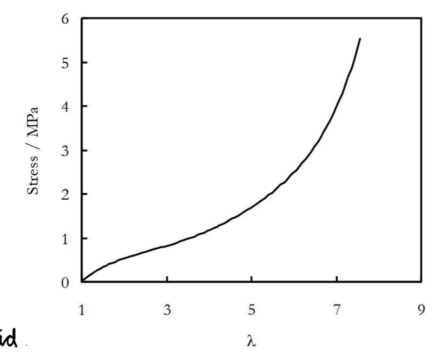
→ Corrosion resistance

→ Low coefficients of friction

## Elastomers

→ Formed by crosslinking polymers.

→ Have many properties of polymers, including:
   - high extensibility
   - Low modulus ($\sim \frac{1}{100000}$ th of metal)




**4 requirements**

→ Temperature above glass transition temperature ($T_g$)

→ Forces between molecules must be weak like in liquid

→ Polymer must not be highly crystalline

→ Polymer must be lightly crosslinked

$$\boxed{\rightarrow \text{Velocity of elastic waves in a material proportional to } \left(\frac{E}{\rho}\right)^{0.5}}$$